Worksheet 4.3 — Solving Exponential & Logarithmic Equations

Grade 12 Advanced Functions (MHF4U) · Unit 4: Exponential and Logarithmic Functions

The Big Ideas

Match bases and set exponents equal, or isolate the power and take logs of both sides; for log equations rewrite in exponential form and check the argument is positive. In every step write the **whole equation**. The solution of $f(x) = c$ is where the curve $y = f(x)$ meets the line $y = c$.

- Same base: $b^f = b^g \Rightarrow f = g$. Write numbers as powers of the same base first.
- Different bases: isolate the power, then $b^x = c \Rightarrow x = \dfrac{\log c}{\log b}$.
- $\log_b x = k \Rightarrow x = b^k$; state the restriction (argument > 0) and reject any solution that breaks it.

Core Concepts

Exponential equations

Take a logarithm of both sides to bring the variable exponent down, then solve the resulting linear equation.

Logarithmic equations

Condense to a single log, rewrite in exponential form, and solve then confirm each argument stays positive.

Watch for extraneous roots

Because logs need positive arguments, discard any solution that makes an argument zero or negative.

Writing solutions

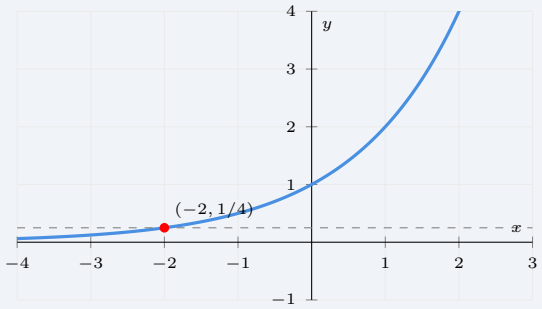
Write the **whole equation** in every step. Isolate the power before taking logs; for a log equation state the restriction (argument > 0) first and check every answer against it. A solution of $f(x) = c$ is where the curve $y = f(x)$ meets the line $y = c$.

Worked Examples

Example 1 — Same base

Solve $2^x = \frac{1}{4}$.

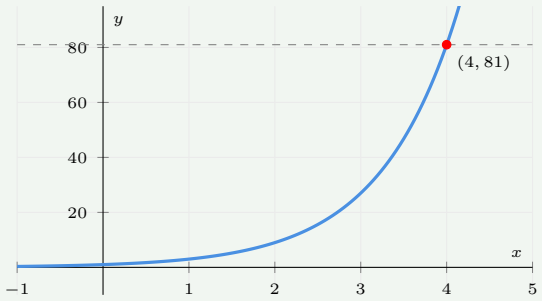
Solution. **Step 1 (common base):** Write $\frac{1}{4} = 2^{-2}$, so $2^x = \frac{1}{4}$ becomes $2^x = 2^{-2}$.
Step 2 (equate the exponents): Both sides have the base 2, so $x = -2$.
Step 3 (check): $2^{-2} = \frac{1}{2^2} = \frac{1}{4}$ ✓. The curve $y = 2^x$ meets $y = \frac{1}{4}$ at $\left(-2, \frac{1}{4}\right)$.
Answer: $x = -2$.



Example 2 — Same base

Solve $3^x = 81$.

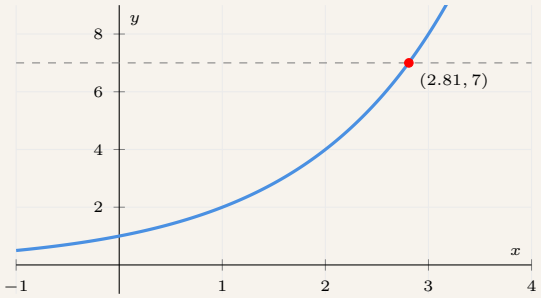
Solution. **Step 1 (common base):** Write $81 = 3^4$, so $3^x = 81$ becomes $3^x = 3^4$.
Step 2 (equate the exponents): $x = 4$.
Step 3 (check): $3^4 = 81$ ✓; the curve $y = 3^x$ meets $y = 81$ at $(4, 81)$.
Answer: $x = 4$.



Example 3 — Take logs

Solve $2^x = 7$.

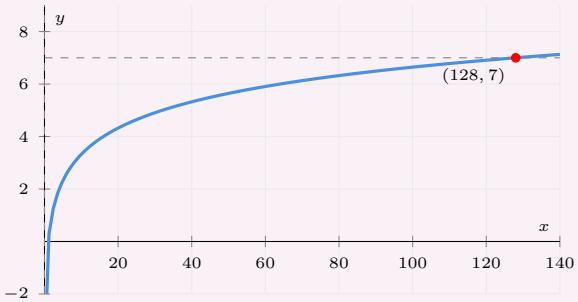
Solution. **Step 1 (why logs):** 7 is not a power of 2, so the bases cannot be matched. Take the logarithm of both sides of $2^x = 7$: $\log(2^x) = \log 7$.
Step 2 (power law): $\log(2^x) = \log 7$ becomes $x \log 2 = \log 7$.
Step 3 (solve): $x = \dfrac{\log 7}{\log 2} \approx \dfrac{0.8451}{0.3010} \approx 2.81$.
Step 4 (check): $2^{2.81} \approx 7$: $2^2 = 4$ and $2^3 = 8$, so an exponent just below 3 is reasonable ✓.
Answer: $x \approx 2.81$.



Example 4 — Log equation

Solve $\log_2 x = 7$.

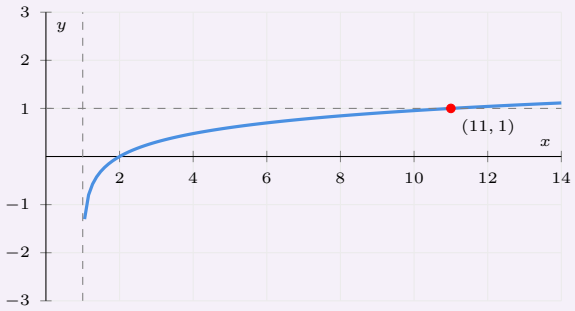
Solution. **Step 1 (restriction):** The argument of $\log_2 x = 7$ must be positive: $x > 0$.
Step 2 (exponential form): Rewrite $\log_2 x = 7$ as $2^7 = x$, so $x = 128$.
Step 3 (check): $128 > 0$ ✓ and $\log_2 128 = 7$ because $2^7 = 128$ ✓.
Answer: $x = 128$.



Example 5 — Log equation

Solve $\log(x-1)=1$.

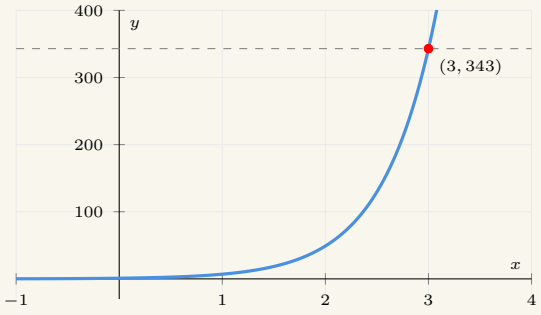
Solution. **Step 1 (restriction):** The argument of $\log(x-1)=1$ must be positive: $x-1>0 \Rightarrow x>1$.
Step 2 (exponential form): Base 10: $\log(x-1)=1$ becomes $x-1=10^1=10$.
Step 3 (solve): $x-1=10 \Rightarrow x=11$.
Step 4 (check): $11>1$ ✓ and $\log(11-1)=\log 10=1$ ✓.
Answer: $x=11$.



Example 6 — Same base

Solve $7^x=343$.

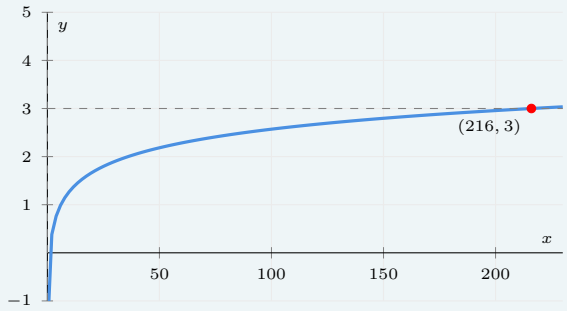
Solution. **Step 1 (common base):** Write $343=7^3$, so $7^x=343$ becomes $7^x=7^3$.
Step 2 (equate the exponents): $x=3$.
Step 3 (check): $7^3=7\cdot 7\cdot 7=343$ ✓.
Answer: $x=3$.



Example 7 — Log equation

Solve $\log_6 x = 3$.

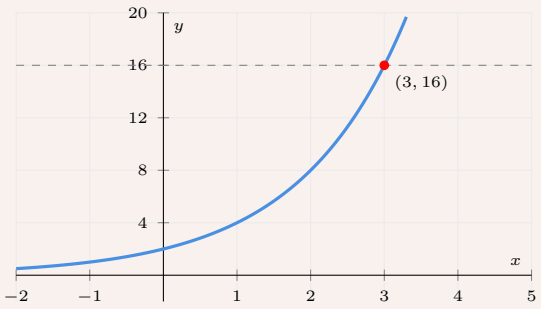
Solution. **Step 1 (restriction):** The argument of $\log_6 x = 3$ must be positive: $x > 0$.
Step 2 (exponential form): Rewrite $\log_6 x = 3$ as $x = 6^3 = 216$.
Step 3 (check): $216 > 0$ ✓ and $\log_6 216 = 3$ because $6^3 = 216$ ✓.
Answer: $x = 216$.



Example 8 — Rewrite first

Solve $2^{x+1} = 16$.

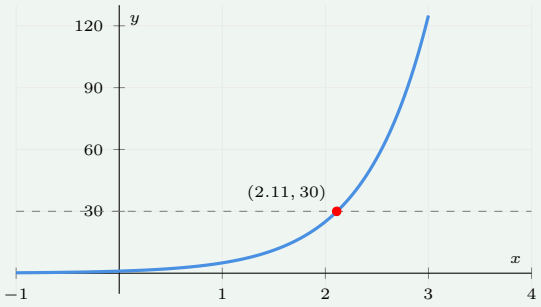
Solution. **Step 1 (common base):** Write $16 = 2^4$, so $2^{x+1} = 16$ becomes $2^{x+1} = 2^4$.
Step 2 (equate the exponents): From $2^{x+1} = 2^4$: $x + 1 = 4$.
Step 3 (solve): $x + 1 = 4 \Rightarrow x = 3$.
Step 4 (check): $2^{3+1} = 2^4 = 16$ ✓.
Answer: $x = 3$.



Example 9 — Take logs

Solve $5^x = 30$.

Solution. **Step 1 (why logs):** 30 is not a power of 5. Take the logarithm of both sides of $5^x = 30$: $\log(5^x) = \log 30$.
Step 2 (power law): $\log(5^x) = \log 30$ becomes $x \log 5 = \log 30$.
Step 3 (solve): $x = \dfrac{\log 30}{\log 5} \approx \dfrac{1.4771}{0.6990} \approx 2.11$.
Step 4 (check): $5^{2.11} \approx 30$: $5^2 = 25$ and $5^3 = 125$ ✓.
Answer: $x \approx 2.11$.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $3^x = 243$.

Question 2

Solve $2^x = 128$.

Question 3

Solve $4^x = 9$ (2 d.p.).

Question 4

Solve $\log_4 x = 3$.

Question 5

Solve $\log(x + 5) = 2$.

Question 6

Solve $4^x = 64$.

Question 7

Solve $2^{x-1} = 32$.

Question 8

Solve $\log_2 x = 6$.

Question 9

Solve $10^x = 500$ (2 d.p.).

Question 10

Solve $\log(x-4)=2$.

Question 11

Solve $3^{2x}=81$.

Question 12

Solve $\log_5 x=2$.

Question 13 — Challenge

Solve $2^x=50$ to two decimal places, showing the log step.

Question Recap & Answer Key

Question Recap

1. Solve $3^x = 243$.
2. Solve $2^x = 128$.
3. Solve $4^x = 9$ (2 d.p.).
4. Solve $\log_4 x = 3$.
5. Solve $\log(x + 5) = 2$.
6. Solve $4^x = 64$.
7. Solve $2^{x-1} = 32$.
8. Solve $\log_2 x = 6$.
9. Solve $10^x = 500$ (2 d.p.).
10. Solve $\log(x - 4) = 2$.
11. Solve $3^{2x} = 81$.
12. Solve $\log_5 x = 2$.
13. Solve $2^x = 50$ to two decimal places, showing the log step.

✓ Answer Key

1. For $3^x = 243$: $243 = 3^5$, so $3^x = 3^5$ and $x = 5$
2. For $2^x = 128$: $128 = 2^7$, so $2^x = 2^7$ and $x = 7$
3. For $4^x = 9$: take logs, $x \log 4 = \log 9$, so $x = \dfrac{\log 9}{\log 4} \approx 1.58$
4. For $\log_4 x = 3$: $x > 0$; exponential form $x = 4^3 = 64$
5. For $\log(x + 5) = 2$: $x + 5 > 0$; exponential form $x + 5 = 10^2 = 100$, so $x = 95$
6. For $4^x = 64$: $64 = 4^3$, so $4^x = 4^3$ and $x = 3$
7. For $2^{x-1} = 32$: $32 = 2^5$, so $x - 1 = 5$ and $x = 6$
8. For $\log_2 x = 6$: $x > 0$; exponential form $x = 2^6 = 64$
9. For $10^x = 500$: take logs, $x = \log 500 \approx 2.70$
10. For $\log(x - 4) = 2$: $x - 4 > 0$; exponential form $x - 4 = 10^2 = 100$, so $x = 104$
11. For $3^{2x} = 81$: $81 = 3^4$, so $3^{2x} = 3^4$, $2x = 4$ and $x = 2$
12. For $\log_5 x = 2$: $x > 0$; exponential form $x = 5^2 = 25$
13. For $2^x = 50$: take logs, $x \log 2 = \log 50$, so $x = \dfrac{\log 50}{\log 2} \approx \dfrac{1.6990}{0.3010} \approx 5.64$